MODEL CARD

Travel Query Intent Classifier

TravelIntent-SVM v1 · TF-IDF + Linear SVM

Symbiosis Artificial Intelligence Institute
Natural Language Processing · NLP-34
Academic year 2026-27 · October 2026

Travel Query Intent Classification Using NLP. Every number on this card comes from one verified run (run ID 87f82a74, seed 42) of the project notebook on the ATIS benchmark, recomputed from stored predictions by Travel NLP Lab Studio.

95.4%
Accuracy
850 test queries

0.683
Macro-F1
95% CI 0.589–0.714

72.1%
Baseline accuracy
always "flight"; Macro-F1 0.042

99.0%
Accuracy kept
after routing the 20% least-certain to a person

1 Model details

Field	Description
Task	Single-label intent classification of English travel queries (22 intents seen in training, e.g. flight, airfare, airline, ground_service, abbreviation)
Model	TF-IDF (word unigrams + bigrams, sublinear tf, 6,772 features) + Linear SVM (LinearSVC, class_weight = balanced, max_iter = 5000)
Model selection	Best of 3 (Logistic Regression, Linear SVM, Multinomial Naive Bayes) by 5-fold stratified CV Macro-F1 on training data only (Linear SVM 0.784 ± 0.033)
Training data	ATIS (Airline Travel Information System) training file: 4,623 queries after removing repeats and train/test overlap (raw: 4,978); "flight" is 74.1%
Test data	ATIS test file: 850 held-out queries (raw: 893), 20 intents; evaluated once
Preprocessing	Lower-casing, whitespace normalisation; repeated queries removed; conflict and overlap checks; stop words kept ("which", "what" carry intent); no stemming
Libraries	Python 3.12.3, scikit-learn 1.8.0, pandas 3.0.2, NumPy 2.4.4; app: Streamlit, Plotly
Developed by	NLP-34 team, SAII — NLP course, October 2026

2 Intended use

- **Intended:** routing short English travel questions (flights, fares, airlines, ground transport, airport and aircraft information) to the right service in a travel chatbot or help desk; classroom demonstration of a classical, explainable NLP pipeline.
- **Out of scope:** bookings, refunds or other binding decisions without human confirmation; non-English or code-mixed queries (e.g. Hinglish); hotel, rail or car-rental questions; intents never seen in training; multi-turn conversations.

3 Overall performance

The model beats the majority-class baseline by 23.3 points of accuracy, and adds most on the less frequent intents, where the baseline scores almost nothing.

Model	Accuracy	Precision (macro)	Recall (macro)	Macro-F1	Weighted-F1
Majority-class baseline	72.1%	—	—	0.042	—
TravelIntent-SVM v1 (TF-IDF + Linear SVM)	**95.4%**	**0.726**	**0.684**	**0.683**	**0.950**

4 Performance across text subgroups

Key audit finding: on long queries (more than 11 words) Macro-F1 falls from 0.849 to 0.438 and accuracy drops 6.1 points versus short queries. Long queries often bundle two requests (flights and fares), which ATIS labels with combined intents that have very few training examples. Treat subgroup Macro-F1 as an indicator, not a precise estimate.

Subgroup	How it was defined	Test n	Accuracy	Macro-F1
Short queries	11 words or fewer (training median)	560	97.5%	0.849
Long queries	**More than 11 words**	**290**	**91.4%**	**0.438**
Questions	Starts with what, which, how, …	255	93.3%	0.807
Request fragments	Any other start ("show me…")	595	96.3%	0.608
Near-duplicates of training	Test query very similar to a training query	18	100.0%	—
All other test queries	No near-duplicate in training	832	95.3%	—

5 Explainability check

Top features: the strongest Linear SVM n-grams are sensible intent cues (airfare: fares, fare, ticket; airline: airlines, what airlines; ground_service: ground transportation), so the model learned real signal.

Word contributions: the classifier is linear, so each intent score is exactly the sum of (TF-IDF × weight) plus a class bias, an exact decomposition, unlike LIME. For "which different airlines go from las vegas to new york city" (true: airline, predicted: flight) the scores differ by only 0.0008; the words push towards airline (airlines −0.72, which −0.36) but the "flight" bias is 0.51 higher. The evidence was right; the preference for the dominant class tipped a near-tie. Most mistakes (32 of 39) have a low margin.

6 Limitations and ethical considerations

- **Rare and unseen intents:** Macro-F1 (0.683) is far below accuracy because "flight" dominates. Four test intents never occur in training (day_name, airfare+flight, flight+airline, flight_no+airline) and can never be predicted; meal (6 training queries) has recall 0.333.
- **Data bias:** ATIS is early-1990s US domestic air travel (US cities and airlines, spoken transcripts); Indian, international, informal or code-mixed queries are likely to be handled worse.
- **No order, negation or context:** TF-IDF sees words and word pairs only, so "not via denver" ≈ "via denver"; typos reduce confidence (no character-level information).
- **Small test set:** many intents have fewer than 5 test queries, so their scores are unreliable; 18 test queries closely match training queries (all correct), and accuracy on the rest is 95.3%.
- **Privacy:** the dataset has no personal data, but a deployed system would handle names and itineraries and would need consent and data-retention rules.

7 Recommendations

- Add examples for rare and combined intents (meal, quantity, ground_fare, flight+airfare) or move to multi-label classification; add an "unknown / other" intent.
- Route low-confidence predictions to a person: flagging the 20% least-certain queries (margin below 1.12) leaves 99.0% accuracy on the rest.
- Try a contextual model (fine-tuned DistilBERT or Sentence-BERT) and character n-grams for typos, then re-run the same subgroup audit.